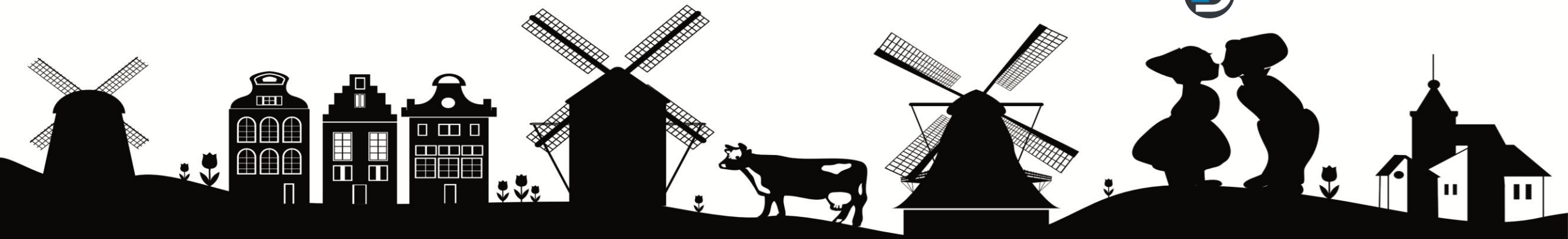


VECTOR in SQL Server 2025: When, Why, and How to Use It

Cláudio Silva

2026-10-10



Abstract

SQL Server 2025 introduces a new data type: VECTOR.

In this session, we will explore what exactly the VECTOR data type is and the problems it helps solve.

I have been doing some proof-of-concept with clients and want to share practical tips, challenges encountered and where these new functionalities fit your scenarios.

Some of the questions we will be answering during this session include:

- What capabilities does it bring that were difficult or inefficient before?
- How does it impact table design?
- Being a new data type, can VECTOR columns be indexed, and how/why might that change the results?

We will also see it in action, how it powers semantic searches by using Similarity search and some more advanced Hybrid searches scenarios, where we combine it with the traditional relational filtering.

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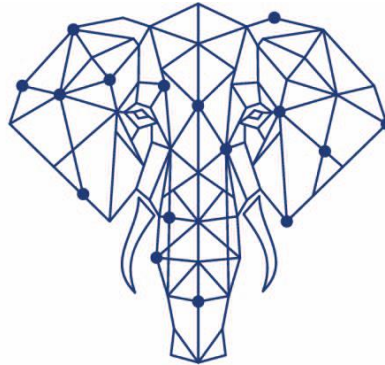
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VECTOR - What is it?

Represents a point in n-dimensional space.

Example: [0.2, 1.5, -3.0]

(VECTOR) EMBEDDING - What is it?

A way to represent data - text/image/documents – as numerical vectors (in the multi-dimensional space) capturing their meaning or properties.

Embeddings that are close to each other are similar in some of their properties, so they can be used for search, clustering, or recommendations

Embedding example

Let's see the word “**dog**” in a 4-dimensional vector

[0.82, 0.65, 0.10, 0.05]

Let's see the word “**airplane**” in a 4-dimensional vector

[0.05, 0.02, 0.91, 0.88]

Let's see the word “**puppy**” in a 4-dimensional vector

[0.79, 0.68, 0.12, 0.04]

Embedding example

Let's see the word “**dog**” in a 4-dimensional vector

Let's see the word “**airplane**” in a 4-dimensional vector

[0.05, 0.02, 0.91, 0.88]

Let's see the word “**puppy**” in a 4-dimensional vector



The diagram illustrates word embeddings in a 4-dimensional space. Three words are listed on the left: 'dog', 'airplane', and 'puppy'. Each word is associated with a 4-dimensional vector. The vector for 'dog' is [0.82, 0.65, 0.10, 0.05], the vector for 'airplane' is [0.05, 0.02, 0.91, 0.88], and the vector for 'puppy' is [0.79, 0.68, 0.12, 0.04]. Arrows point from each word to its corresponding vector. The vectors for 'dog' and 'puppy' are highlighted with orange boxes around each element.

[0.82, 0.65, 0.10, 0.05]
[0.79, 0.68, 0.12, 0.04]

Who am I?

Cláudio Silva

(He / Him)

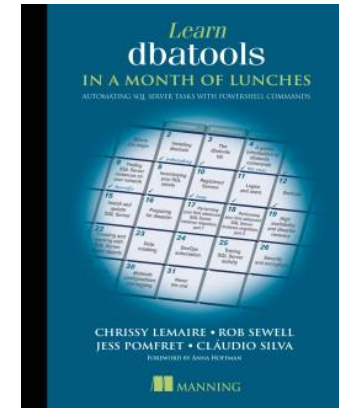
What do I do?



- Performance Tuning
- Automation

Open-Source Contributor

- dbatools
- dbachecks



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VECTOR and SQL Server

Available on: SQL Server 2025 + Azure Flavours + SQL Database in Microsoft Fabric

- VECTOR(x)
 - X being the number of dimensions (can't be more than 1998)
 - Check the model you are using
 - Single-precision (4-byte) floating-point value (Float(32))



Why use it with(in) SQL Server?

```
where
    b.city = 'Seattle'
and
    @p.STDistance(b.geo_location) < 1000 -- 1 km
and
    r.stars >= 4
and
    b.reviews >= 30
and
    json_contains(b.custom_attributes, cast(1 as bit), '$.local_recommended') = 1
and
    regexp_like(category, '(:new york|detroit) style')
and
    vector_distance('cosine', re.embedding, @e) < 0.2
```

Source: [SQL Server 2025 Vector and AI features released in RTM - Azure SQL Dev Corner](#)



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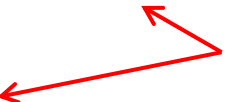
DEMOS

Improvements

Since the SQL Server 2025 box product release:

- Float(16) – Half-precision = half-size
 - Most of the time, identical results (do your testing)
 - (on premises) You need to turn on [PREVIEW_FEATURES](#) (a database-scoped configuration)
- No longer read-only after adding a vector index
- Improvements to the similarity search

THESE ARE NOT AVAILABLE ON PREMISES!



Vector index – Limitations in earlier versions

- **Post-filtering** only:
 - Predicates are applied only after vector retrieval, not during the search process. This can result in fewer rows returned than expected when filters are applied.
- **Read-only tables:**
 - Tables with vector indexes are read-only.
- **Manual TOP_N tuning:**
 - You must manually adjust the TOP_N parameter in VECTOR_SEARCH to compensate for post-filtering, often requiring oversized values to get the desired number of results.
- **Primary Key Clustered requirement:**
 - The table must have a primary key clustered index on a single INT column.

Source: [CREATE VECTOR INDEX \(Transact-SQL\) - Earlier vector index version limitations](#)

Vector index – Limitations in current versions

- Vector indexes **can't be partitioned**. No partition support.
- The table must have a primary key clustered index.
- Vector indexes aren't replicated to subscribers.
- Tables with vector indexes can't be truncated using TRUNCATE TABLE.
- Vector indexes can't be deployed with DacPac or BACPAC.
- To create a VECTOR index, we need at least 100 rows.

Source: [CREATE VECTOR INDEX \(Transact-SQL\) - Current limitations \(applies to the latest version too\)](#)

Suggestions for the application side

Make sure you are using an updated driver that supports the VECTOR data type.

Examples:

- Microsoft.Data.SqlClient 6.1.0 or later
- Microsoft JDBC Driver for SQL Server 13.1.0 and later

If you want to do some bulk insert/update with embeddings, do it from the application side will give you more control.

Think about:

- Rate limit
- Error control
- Retries

What else?

We have just used the 'cosine' metric.

There are two others:

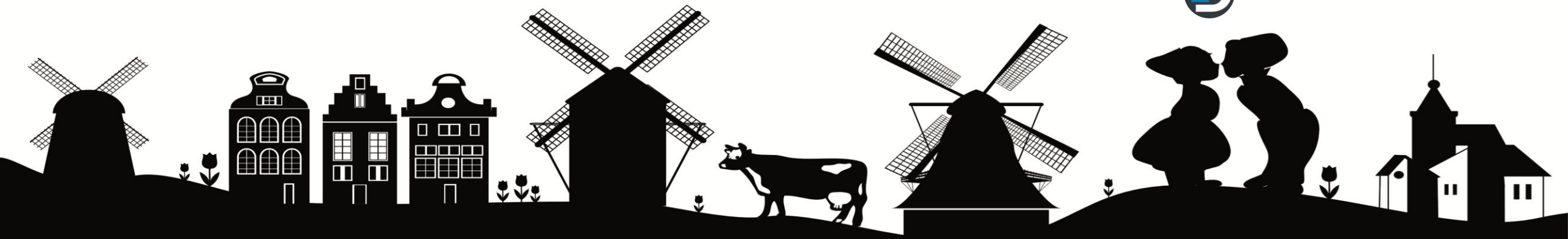
- 'euclidean'
- 'dot'

Check Andrew Pruski's post: [Distance Metrics for Semantic Similarity Searches in SQL Server 2025 | DBA From The Cold](#)

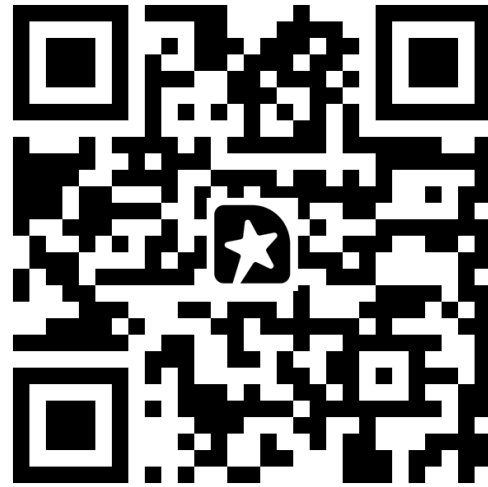
We saw VECTOR_DISTANCE and VECTOR_SEARCH. There are a couple more VECTOR functions:

- [VECTOR_NORM](#): Returns the measure of its length or magnitude
- [VECTOR_NORMALIZE](#): Will normalize a vector, which is a vector scaled to have a length of 1 in a given norm type
- [VECTORPROPERTY](#): Returns specific properties of a given vector

Questions?



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